

Homework 3: Applications to Differential Equations

BCS A, E, F, G

Problem 1. *The Problem of Distinct Real Eigenvalues.*

Solve

$$\frac{dx}{dt} = 2x + 3y, \quad \frac{dy}{dt} = 2x + y.$$

Problem 2. *The Problem of Repeated Real Eigenvalues.*

Solve

$$\mathbf{X}' = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \mathbf{X}.$$

Problem 3. *The Problem of Eigenvalue of Multiplicity Three.*

Solve

$$\mathbf{X}' = \begin{bmatrix} 2 & 1 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix} \mathbf{X}.$$

Problem 4. *The Problem of Complex Eigenvalues.*

Solve

$$\frac{dx}{dt} = 6x - y, \quad \frac{dy}{dt} = 5x + 4y.$$

Problem 5. *The Problem of Real Solutions Corresponding to a Complex Eigenvalue.*

Solve the initial-value problem

$$\mathbf{X}' = \begin{bmatrix} 2 & 8 \\ -1 & -2 \end{bmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Submission Date: October 5th, 2024 (11:59PM)

Good Luck